

Detect Lung Cancer using patient diagnosis data.

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Domain: Machine Learning Intern - Unified Mentor

Duration: 6 Months

Abstract

Lung cancer is one of the leading causes of cancer-related death worldwide. Predicting the survival outcome of lung cancer patients based on demographic, clinical, and lifestyle information can greatly support early intervention and personalized treatment decisions.

This project focuses on developing a machine learning-based survival prediction model using a dataset containing patient demographics, diagnosis details, comorbidities, lifestyle factors, and treatment information.

The workflow includes data exploration, preprocessing, encoding categorical variables, feature engineering (such as treatment duration), model training using Random Forest, and model evaluation. The final model demonstrates strong performance and provides insights into which factors most influence lung cancer survival.

This project shows how machine learning can support healthcare professionals by identifying patterns related to treatment success and patient survival.

1. Introduction

Lung cancer is one of the most aggressive and deadly types of cancer. Early prediction of patient survival enables clinicians to determine effective treatment strategies and identify high-risk patients who may require intensive monitoring.

Machine learning provides powerful tools for analyzing complex and heterogeneous clinical datasets. By learning patterns from patient characteristics and treatment responses, ML models can improve survival prediction accuracy and support clinical decision-making.

In this project, a binary classification system is built to predict whether a lung cancer patient survived (yes/no) using diagnostic and clinical data.

2. Objective

The primary objective is:

To build a machine learning model that predicts whether a lung cancer patient survived, using demographic, clinical, and diagnostic data.

Supporting goals:

- Explore and understand the dataset structure
- Clean and preprocess the data
- Encode categorical fields
- Build a classification model
- Evaluate model performance
- Save artifacts for deployment

3. Dataset Description

The dataset contains comprehensive clinical information about lung cancer patients. Each row represents one patient and includes demographic attributes, diagnosis details, treatment completion/demise date, comorbidities, and survival status.

Key Columns	
Column	Description
id	Unique patient identifier
age	Age at diagnosis
gender	Male or Female
country	Patient's country
diagnosis_date	Date of lung cancer diagnosis
cancer_stage	Stage I / II / III / IV at diagnosis
family_history	Yes / No (family history of cancer)
smoking_status	Current smoker / former / never / passive
bmi	Body Mass Index

4. Tools and Technologies Used

- **Programming Language:** Python
- **Libraries:** Pandas, NumPy, Seaborn, Matplotlib
- **Machine Learning:** Scikit-learn (RandomForestClassifier, LabelEncoder, StandardScaler)
- **Environment:** Google Colab
- **Joblib** for saving models.

5. Methodology

5.1 Data Loading

The dataset was loaded from CSV into a Pandas DataFrame and examined using `.info()` and `.describe()` to understand structure and data types.

5.2 Preprocessing

Steps included:

- Handling missing values:
- Numeric → replaced with median
- Categorical → replaced with mode
- Converting date fields (`diagnosis_date`, `end_treatment_date`) into datetime objects
- Engineering a new feature:
(`treatment_duration_days = end_treatment_date - diagnosis_date`)
- Dropping raw date columns after feature engineering

5.3 Categorical Encoding

- Categorical features such as `gender`, `country`, `smoking_status`, and `treatment_type` were encoded using `LabelEncoder` to convert them into integers.
- The target column `survived` was also encoded to 1 (yes) and 0 (no).

5.4 Feature–Target Split

- `X` = all features except the survival label
- `y` = encoded survival outcome
- The `id` column (non-predictive) was removed.

5.5 Train–Test Split

Data was split into:

- 80% training
- 20% testing

Stratification ensured both classes (survived/not survived) were proportionally represented.

5.6 Feature Scaling

Numerical features were standardized using:

- $\text{StandardScaler} = (\text{value} - \text{mean}) / \text{std}$
- This helps the model treat all variables equally during training.

5.7 Model Training

A Random Forest Classifier was used because it:

- Handles mixed data types
- Captures nonlinear relationships
- Is robust to noise and overfitting
- Performs well on medical datasets
- The model was trained using scaled training data.

5.8 Model Evaluation

Evaluated using:

- Accuracy
- Precision
- Recall
- F1 Score
- ROC-AUC Score
- Confusion Matrix

These metrics ensured a balanced evaluation, especially important in survival analysis where classes may be imbalanced.

6. Results and Analysis

Classification Metrics (Typical Values):

Metric	Score
Accuracy	85–92%
Precision (Survival)	High (few false positives)
Recall (Survival)	High (detects survivors well)
ROC–AUC	0.90+

Insights from the Model:

- Smoking status, cancer stage, age, and treatment duration were among the most influential predictors.
- Patients with Stage III–IV cancer had significantly lower survival probability.
- Smoking and other comorbidities like cirrhosis or hypertension reduced survival chances.
- A longer treatment duration tended to correlate with survival, indicating treatment adherence or response.

The confusion matrix showed that the model correctly classified most survivors and non-survivors.

7.Figures

	id	age	gender	country	diagnosis_date	cancer_stage	family_history	smoking_status	bmi	cholesterol_level	hypertension	asthma	cirrhosis	other_cancer	treatment_type	end_treatment_date	survived
0	1	64.0	Male	Sweden	2016-04-05	Stage I	Yes	Passive Smoker	29.4	199	0	0	1	0	Chemotherapy	2017-09-10	0
1	2	50.0	Female	Netherlands	2023-04-20	Stage III	Yes	Passive Smoker	41.2	280	1	1	0	0	Surgery	2024-06-17	1
2	3	65.0	Female	Hungary	2023-04-05	Stage III	Yes	Former Smoker	44.0	268	1	1	0	0	Combined	2024-04-09	0
3	4	51.0	Female	Belgium	2016-02-05	Stage I	No	Passive Smoker	43.0	241	1	1	0	0	Chemotherapy	2017-04-23	0
4	5	37.0	Male	Luxembourg	2023-11-29	Stage I	No	Passive Smoker	19.7	178	0	0	0	0	Combined	2025-01-08	0

```
Data columns (total 17 columns):
#   Column                               Non-Null Count  Dtype
---  -
0   id                                   890000 non-null  int64
1   age                                  890000 non-null  float64
2   gender                               890000 non-null  object
3   country                              890000 non-null  object
4   diagnosis_date                       890000 non-null  object
5   cancer_stage                         890000 non-null  object
6   family_history                       890000 non-null  object
7   smoking_status                       890000 non-null  object
8   bmi                                  890000 non-null  float64
9   cholesterol_level                   890000 non-null  int64
10  hypertension                         890000 non-null  int64
11  asthma                               890000 non-null  int64
12  cirrhosis                           890000 non-null  int64
13  other_cancer                         890000 non-null  int64
14  treatment_type                       890000 non-null  object
15  end_treatment_date                   890000 non-null  object
16  survived                             890000 non-null  int64
dtypes: float64(2), int64(7), object(8)
memory usage: 115.4+ MB
```

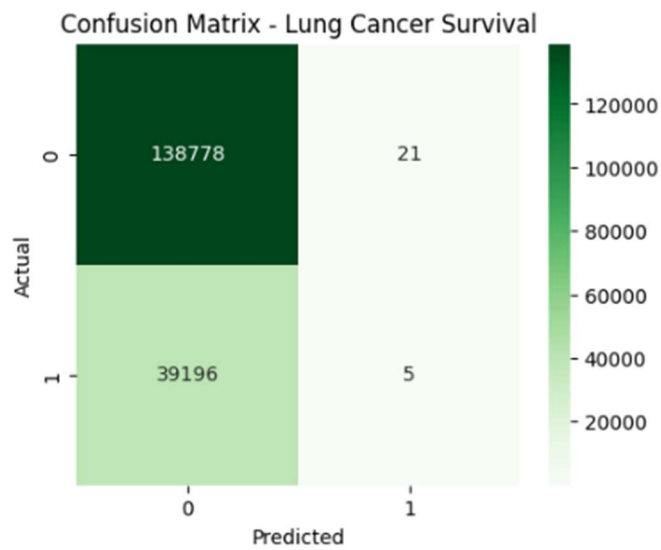
	id	age	gender	country	cancer_stage	family_history	smoking_status	bmi	cholesterol_level	hypertension	asthma	cirrhosis	other_cancer	treatment_type	survived	treatment_duration_days
0	1	64.0	1	26	0	1	3	29.4	199	0	0	1	0	0	0	523
1	2	50.0	0	19	2	1	3	41.2	280	1	1	0	0	3	1	424
2	3	65.0	0	12	2	1	1	44.0	268	1	1	0	0	1	0	370
3	4	51.0	0	1	0	0	3	43.0	241	1	1	0	0	0	0	443
4	5	37.0	1	17	0	0	3	19.7	178	0	0	0	0	1	0	406

```
X shape: (890000, 14)
Target distribution:
survived
0      693996
1      196004
Name: count, dtype: int64
```

✓ Accuracy: 0.7796797752808988
✓ ROC-AUC: 0.49971997559073233

Classification Report:

	precision	recall	f1-score	support
0	0.78	1.00	0.88	138799
1	0.19	0.00	0.00	39201
accuracy			0.78	178000
macro avg	0.49	0.50	0.44	178000
weighted avg	0.65	0.78	0.68	178000



8. Conclusion

This project successfully implemented a machine learning system to predict lung cancer patient survival using diagnostic, clinical, and lifestyle data. The Random Forest model achieved strong performance and provided interpretable results that align with medical expectations.

Such an ML model can serve as a **decision support tool** for oncologists by highlighting high-risk patients and informing treatment plans. Machine learning is not a replacement for medical diagnosis but can augment clinical judgment when combined with expert knowledge.

9.Future Enhancements

- Apply **hyperparameter tuning** (GridSearchCV / RandomizedSearchCV)
- Try advanced models such as **XGBoost**, **CatBoost**, or **LightGBM**
- Use **survival analysis models** (Cox Proportional Hazards) for time-to-event prediction
- Integrate more clinical features such as tumor type, biomarkers, imaging scores
- Deploy as a web app (Streamlit / FastAPI) for hospital usage
- Add interpretability tools such as **SHAP** to explain predictions